Phase 2 · JavaScript — JS Core, DOM, Async & APIs

Assignment No. 2

Title: Price & Discount Calculator — JavaScript Fundamentals

Session: Week 5 — Session 15

Max Marks: 10

Objective:

The purpose of this assignment is to test your understanding of JavaScript variables, operators, conditionals, functions, and scope — the core concepts covered in Sessions 12 to 15 — by building a practical price and discount calculator using pure JavaScript with no DOM or arrays.

Assignment: Build a price and discount calculator using JavaScript functions and conditionals.

Requirements:

- Declare variables for a product: name (string), basePrice (number), discountPercent (number), taxRate (number), and stockStatus (string — 'in_stock', 'limited', or 'out_of_stock').
- Write a calculateDiscountedPrice(price, discount) function — returns the price after applying the discount percentage.
- Write a calculateTax(price, taxRate) function — returns the tax amount (taxRate % of price).
- Write a calculateFinalPrice(price, discount, taxRate) function — returns the final price: discounted price plus tax amount.
- Write an isFreeShipping(finalPrice) function — returns 'Free Shipping' if finalPrice >= 500, otherwise returns 'Shipping: Rs.49'.
- Write a getStockMessage(status) function using if/else or switch — returns 'In Stock', 'Limited Stock', or 'Out of Stock' based on the status string.
- Call all six functions with your product variables and print a full price summary to the browser console using console.log.

Breakdown of Marks (10):

Criteria	Marks
Variable declarations with correct product data	1
calculateDiscountedPrice and calculateTax functions — correct logic	3
calculateFinalPrice combining both functions correctly	2
isFreeShipping and getStockMessage with correct conditionals	2
Full price summary printed to console with all six values **Total**	2 **10**

Expected Output:

A JavaScript file with all six functions working correctly — the complete price summary (discounted price, tax, final price, shipping, and stock status) printed to the browser console.

Assignment Submission Guidelines

1. Create a new GitHub repository for each assignment with a meaningful repository name.
2. Upload your complete project to GitHub and ensure the repository is accessible.
3. Create a Google Doc containing:
    - Learner Name
    - Batch Code
    - Assignment Number & Assignment Name
    - GitHub Repository Link
4. Set the Google Doc sharing permission to Anyone with the link can view.
5. Submit the Google Doc link through the assignment submission portal. 6. Ensure your project is complete, runs without errors, and includes all required features.
7. Submit before the deadline. Incomplete or inaccessible submissions may not be evaluated.